Lecture 30: Markov chain Monte Carlo (MCMC) III

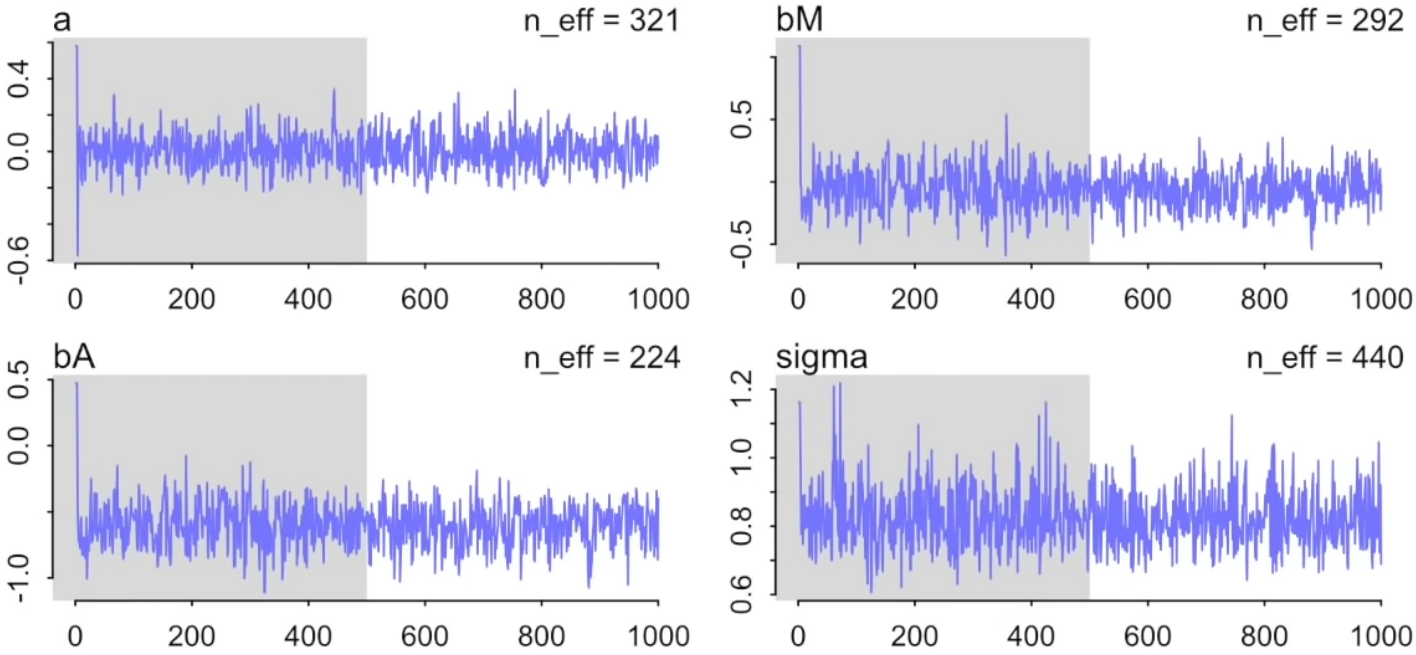
Good chains, bad chains and diagnostics

Some HMC practice with `ulam()`

- Finish the `Divorce` example
- Revisiting the terrain ruggedness example (SR book)
- Re-approximate the posterior
- Practical details II
- What to expect from a healthy Markov chain? Recognise a wounded one and fix it.

Trace plot II

The "hairy caterpillar"



How many chains?

- Need more than one chain to check convergence
- *Convergence*: each chain explores the same distribution and each chain explores it well.

$$
\begin{aligned}
D_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta_M M_i + \beta_A A_i \\
\alpha &\sim \text{Normal}(0,0.2) \\
\beta_M &\sim \text{Normal}(0,0.5) \\
\beta_A &\sim \text{Normal}(0,0.5) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

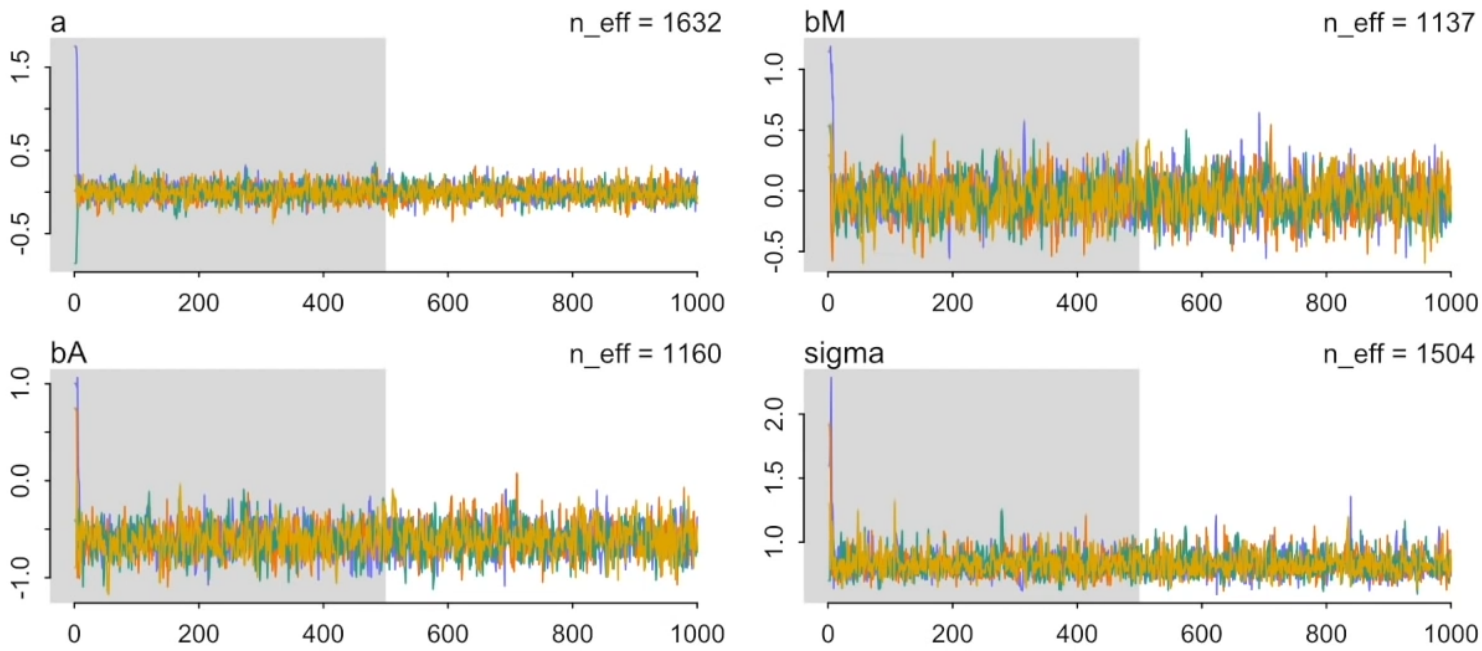
```
library(cmdstanr)
mHMC <- ulam( f , data=dat )

mHMC <- ulam( f , data=dat , chains=4 , cores=4 )
```

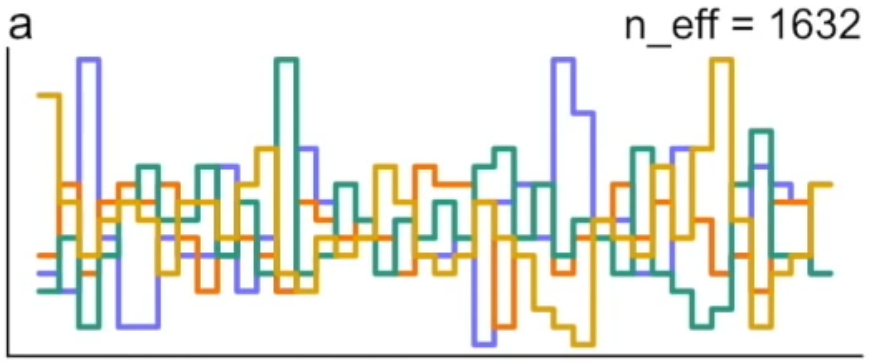
Summary: one chain for debugging, several (4 is OK) chains to decide if chains are valid and final run for inference: one (long) chain (or several in parallel and merge non warm-up samples)

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Trace plot III



Trace rank ("Trank") plots

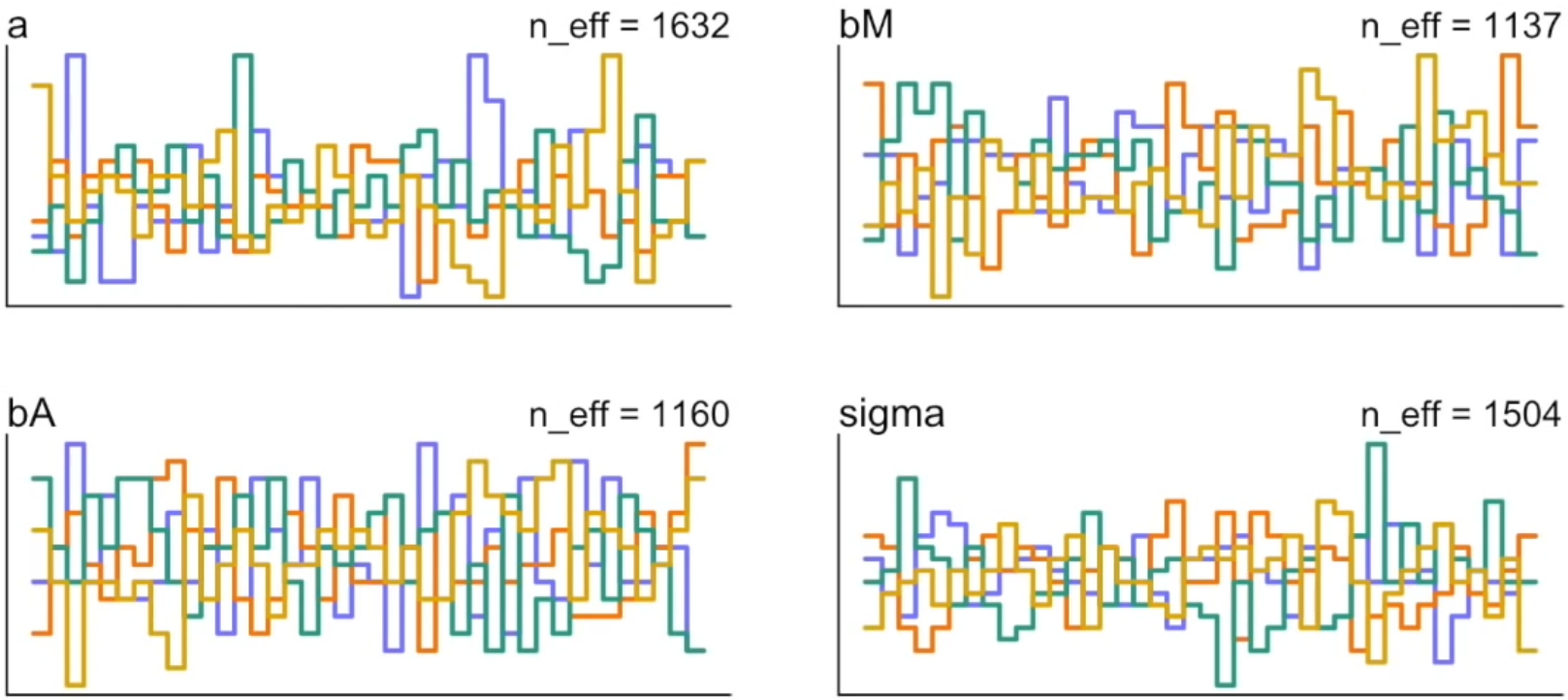


(from several chains)

Instead of parameter values on the y-axis, it's been converted to ranks.

Quick heuristic way to check mixing in the chain and whether a chain is above or below the others.

Trank plots



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R-hat $\hat{R}$

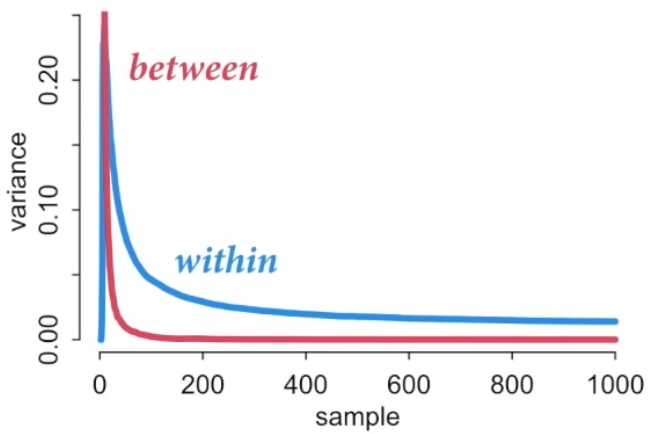
When a chain is in convergence mode

- The start and end of each chain explores the same region
- Independent chain explores the same region

R-hat is a ratio of variances (total divided by average within): as the total variance shrinks to average variance within chains, R-hat approaches one.

Note: there is NO guarantee, it is not a test. Rather a warning towards danger, not a safety indicator.

```
> precis(mHMC)
       mean   sd  5.5% 94.5% n_eff Rhat4
a      0.00 0.10 -0.16  0.16  1632     1
bM    -0.06 0.17 -0.32  0.21  1137     1
bA    -0.61 0.17 -0.86 -0.34  1160     1
sigma  0.83 0.09  0.70  0.99  1504     1
>
```



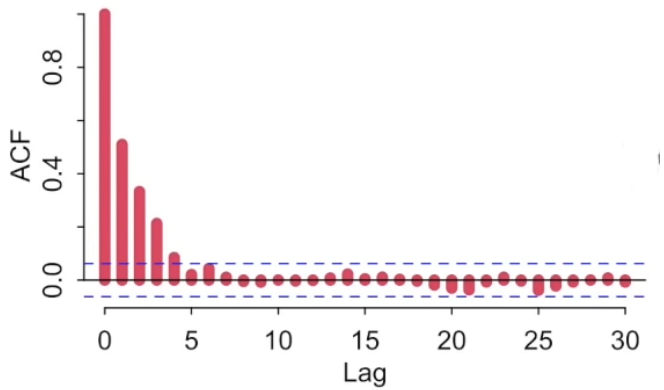
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N-eff

Number of effective parameters

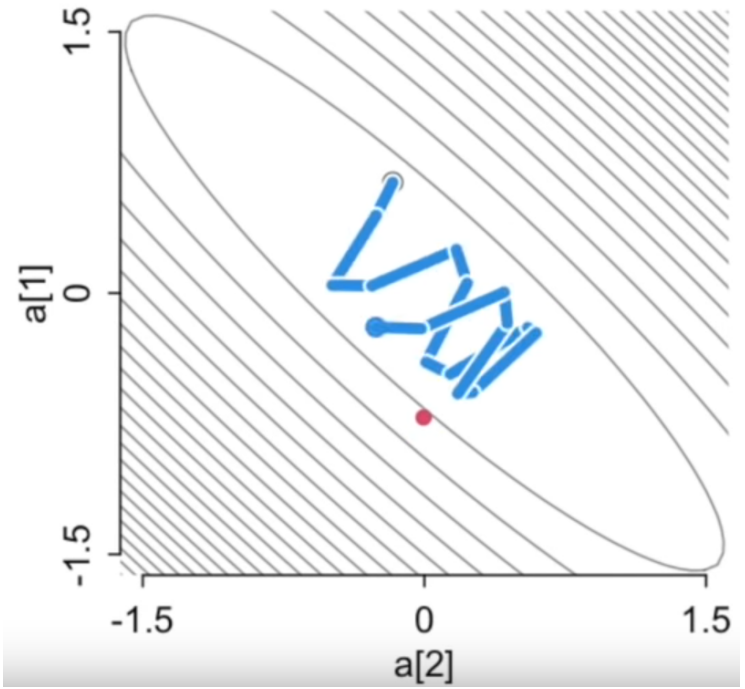
- Estimates the number of *effective samples*.
- "How long would the chain be, if each sample was independent of the one before it? "
- When samples are auto-correlated, you have less effective samples.

```
> precis(mHMC)
       mean   sd  5.5% 94.5% n_eff Rhat4
a      0.00 0.10 -0.16  0.16  1632     1
bM    -0.06 0.17 -0.32  0.21  1137     1
bA    -0.61 0.17 -0.86 -0.34  1160     1
sigma  0.83 0.09  0.70  0.99  1504     1
>
```



Divergent transitions

- Divergent transition: some sort or a rejected proposal
- Notice the long (discrete) segments
- Simulations diverge from the true path
- Many DTs: could indicate poor exploration and possible bias
- A powerful tool to decide if exploration is adequate...in more complex models
- There are ways to program around



Back to the terrain ruggedness example

`quap()` approximation

R code 9.10

```
m8.5 <- quap(
    alist(
        log_gdp_std ~ dnorm( mu , sigma ) ,
        mu <- a[cid] + b[cid]*( rugged_std - 0.215 ) ,
        a[cid] ~ dnorm( 1 , 0.1 ) ,
        b[cid] ~ dnorm( 0 , 0.3 ) ,
        sigma ~ dexp( 1 )
    ) ,
    data=dd )
precis( m8.5 , depth=2 )
```

```
        mean   sd  5.5% 94.5%
a[1]    0.89 0.02  0.86  0.91
a[2]    1.05 0.01  1.03  1.07
b[1]    0.13 0.07  0.01  0.25
b[2]   -0.14 0.05 -0.23 -0.06
sigma   0.11 0.01  0.10  0.12
```

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Back to the terrain ruggedness example

Now the `ulam()` version

```
dat_slim <- list(
    log_gpd_std = dd$log_gdp_std,
    rugged_std = dd$rugged_std,
    cid = as.integer( dd$cid )
)

m9.1 <- ulam(
    alist(
        log_gdp_std ~ dnorm( mu , sigma ) ,
        mu <- a[cid] + b[cid]*( rugged_std - 0.215 ) ,
        a[cid] ~ dnorm( 1 , 0.1 ) ,
        b[cid] ~ dnorm( 0 , 0.3 ) ,
        sigma ~ dexp( 1 )
    ) ,
    data=dat_slim , chains=4 , cores=4 , iter=1000 )
```

R code
9.14

Back to the terrain ruggedness example

Hamiltonian flows

```
    data=dat_slim , chains=4 , cores=4 , iter=1000 )
```

```
show( m9.1 )
```

```
Hamiltonian Monte Carlo approximation
2000 samples from 4 chains

Sampling durations (seconds):
         warmup sample total
chain:1   0.12   0.08  0.20
chain:2   0.12   0.08  0.19
chain:3   0.12   0.08  0.20
chain:4   0.12   0.08  0.20
```

- Number of samples = total minus warm-up
- Default warm-up is 1/2 samples
- Check values are the same as with `quap()` (R code 9,18 in SR book)

Badly behaving chains

Few data and extremely flat priors

```
y <- c(-1,1)
set.seed(11)
m9.2 <- ulam(
    alist(
        y ~ dnorm( mu , sigma ),
        mu <- alpha,
        alpha ~ dnorm( 0 , 1000 ),
        sigma ~ dexp( 0.0001 )
    ) , data=list(y=y) , chains=3 , cores=3 )
```

Badly behaving chains

Few data and extremely flat priors

```
y <- c(-1,1)
set.seed(11)
m9.2 <- ulam(
    alist(
        y ~ dnorm( mu , sigma ),
        mu <- alpha,
        alpha ~ dnorm( 0 , 1000 ),
        sigma ~ dexp( 0.0001 )
    ) , data=list(y=y) , chains=3 , cores=3 )
```

```
> precis(m9.2)
          mean     sd    5.5%   94.5% n_eff Rhat4
alpha -27.98 280.33 -485.11  291.74   121  1.01
sigma 390.82 993.07    1.83 1870.86   169  1.03
>
```

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Badly behaving chains

Few data and extremely flat priors

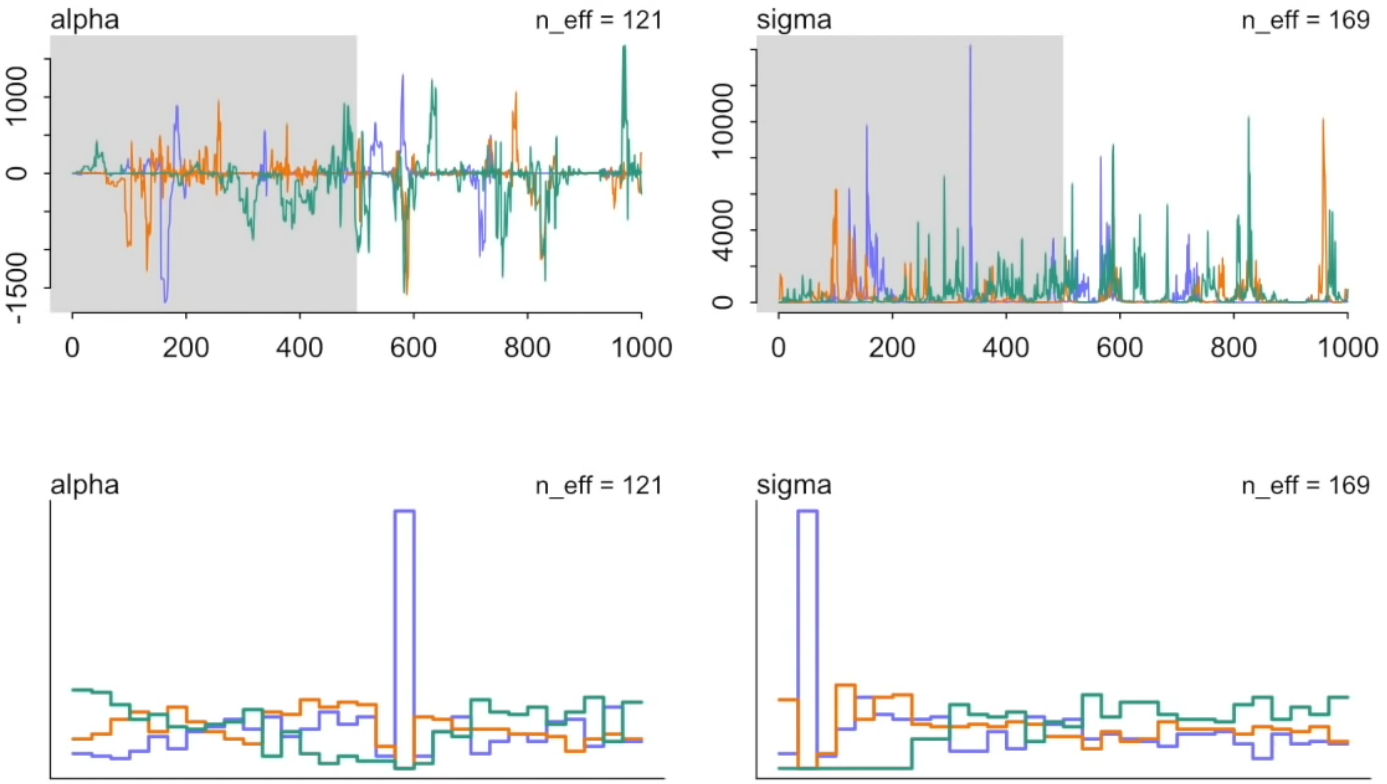
```
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set.seed(11)
m9.2 <- ulam(
    alist(
        y ~ dnorm( mu , sigma ),
        mu <- alpha,
        alpha ~ dnorm( 0 , 1000 ),
        sigma ~ dexp( 0.0001 )
    ) , data=list(y=y) , chains=3 , cores=3 )
```

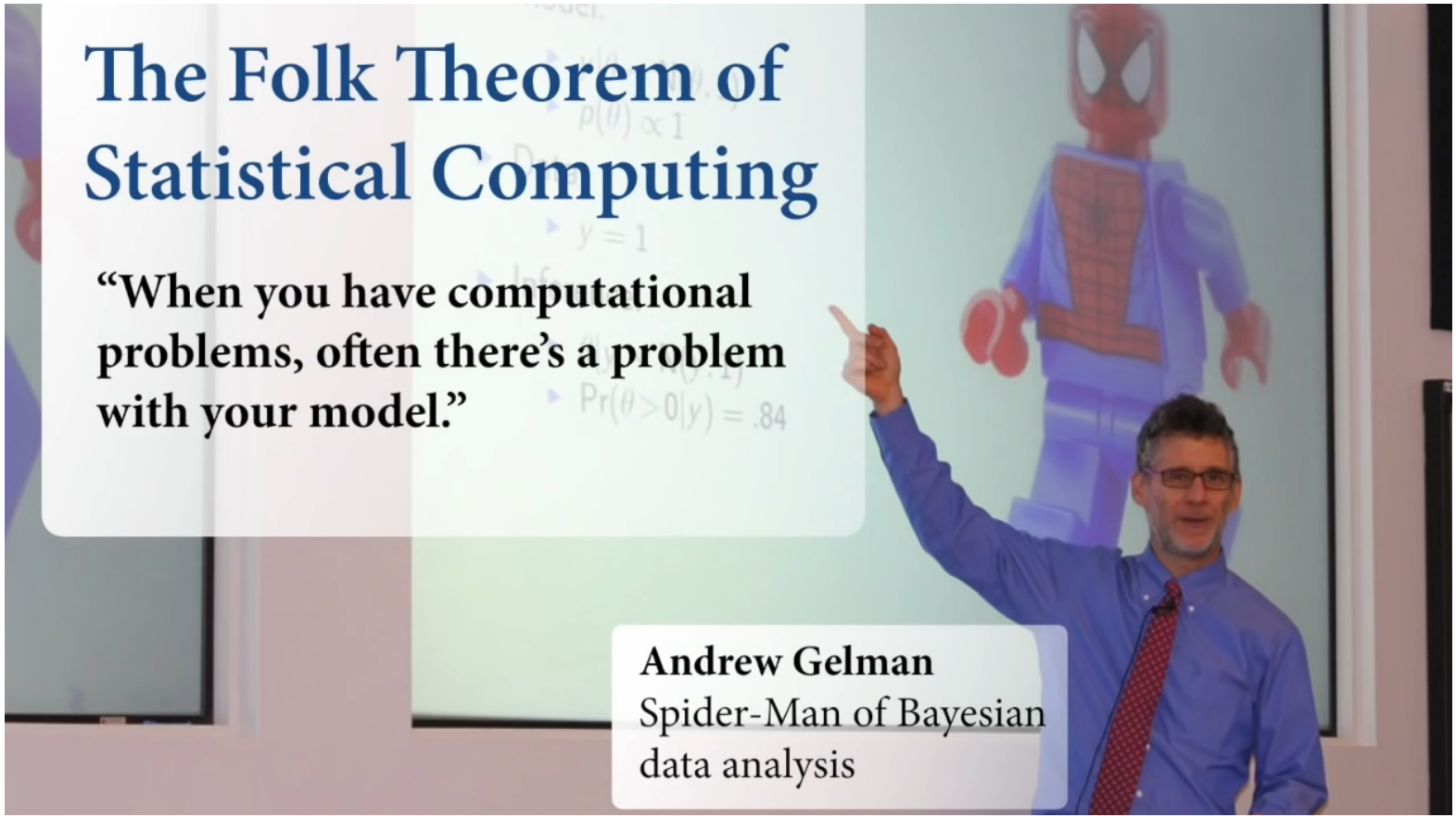
```
> precis(m9.2)
          mean     sd    5.5%   94.5% n_eff Rhat4
alpha -27.98 280.33 -485.11  291.74   121  1.01
sigma 390.82 993.07    1.83 1870.86   169  1.03
>
```

```
Warning: 108 of 1500 (7.0%) transitions ended with a divergence.
This may indicate insufficient exploration of the posterior distribution.
Possible remedies include:
  * Increasing adapt_delta closer to 1 (default is 0.8)
  * Reparameterizing the model (e.g. using a non-centered parameterization)
  * Using informative or weakly informative prior distributions
```

Badly behaving chains

Trace and trank plots




The Folk Theorem of Statistical Computing
"When you have computational problems, often there's a problem with your model."
Andrew Gelman
Spider-Man of Bayesian data analysis

Badly behaving chains

Weakly informative priors save the day

```
y <- c(-1,1)
set.seed(11)
m9.2 <- ulam(
    alist(
        y ~ dnorm( mu , sigma ),
        mu <- alpha,
        alpha ~ dnorm( 0 , 1000 ),
        sigma ~ dexp( 0.0001 )
    ) , data=list(y=y) , chains=3 ,
cores=3 )
```

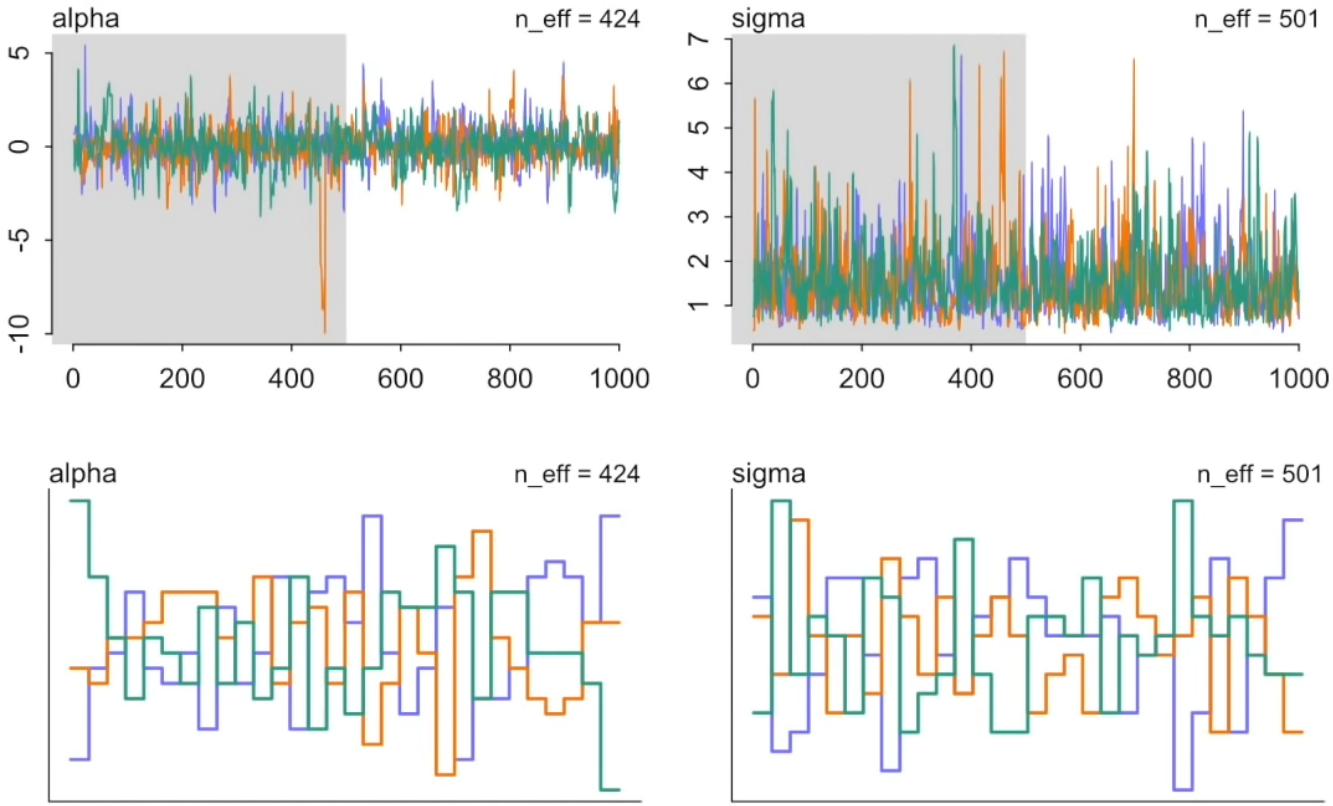
```
> precis(m9.2)
          mean      sd    5.5%   94.5% n_eff Rhat4
alpha -27.98 280.33 -485.11  291.74   121  1.01
sigma 390.82 993.07    1.83 1870.86   169  1.03
>
```

```
m9.3 <- ulam(
    alist(
        y ~ dnorm( mu , sigma ),
        mu <- alpha,
        alpha ~ dnorm( 1 , 10 ),
        sigma ~ dexp( 1 )
    ) , data=list(y=y) , chains=3 , cores=3 )
```

```
> precis(m9.3)
      mean   sd  5.5% 94.5% n_eff Rhat4
alpha 0.06 1.07 -1.62  1.80   424  1.01
sigma 1.51 0.80  0.65  3.02   501  1.00
>
```

Badly behaving chains

Trace and trank plots: the sick Markov chain is healed



With two observations only, the weakly informative priors have no effect on inferences...but to make them possible.

Non-identifiable problems

Another "Folk theorem of computational statistics" situation

$$y_i \sim \text{Normal}(\mu, \sigma)$$
$$\mu = \alpha_1 + \alpha_2$$
$$\sigma \sim \text{Exponential}(1)$$

R code 9.22

```
set.seed(41)
y <- rnorm( 100 , mean=0 , sd=1 )
```

R code 9.23

```
m9.4 <- ulam(
    alist(
        y ~ dnorm( mu , sigma ) ,
        mu <- a1 + a2 ,
        a1 ~ dnorm( 0 , 1000 ),
        a2 ~ dnorm( 0 , 1000 ),
        sigma ~ dexp( 1 )
    ) ,
    data=list(y=y) , chains=2 )
precis( m9.4 )
```

```
        mean     sd    5.5%  94.5% n_eff Rhat
sigma   1.05   0.08    0.91   1.17     5 1.32
a2     21.46 291.97 -473.77 475.59     2 2.01
a1    -21.27 291.96 -475.54 473.96     2 2.01
```

Non-identifiable problems

Typical patterns...again, solved by weakly informative priors

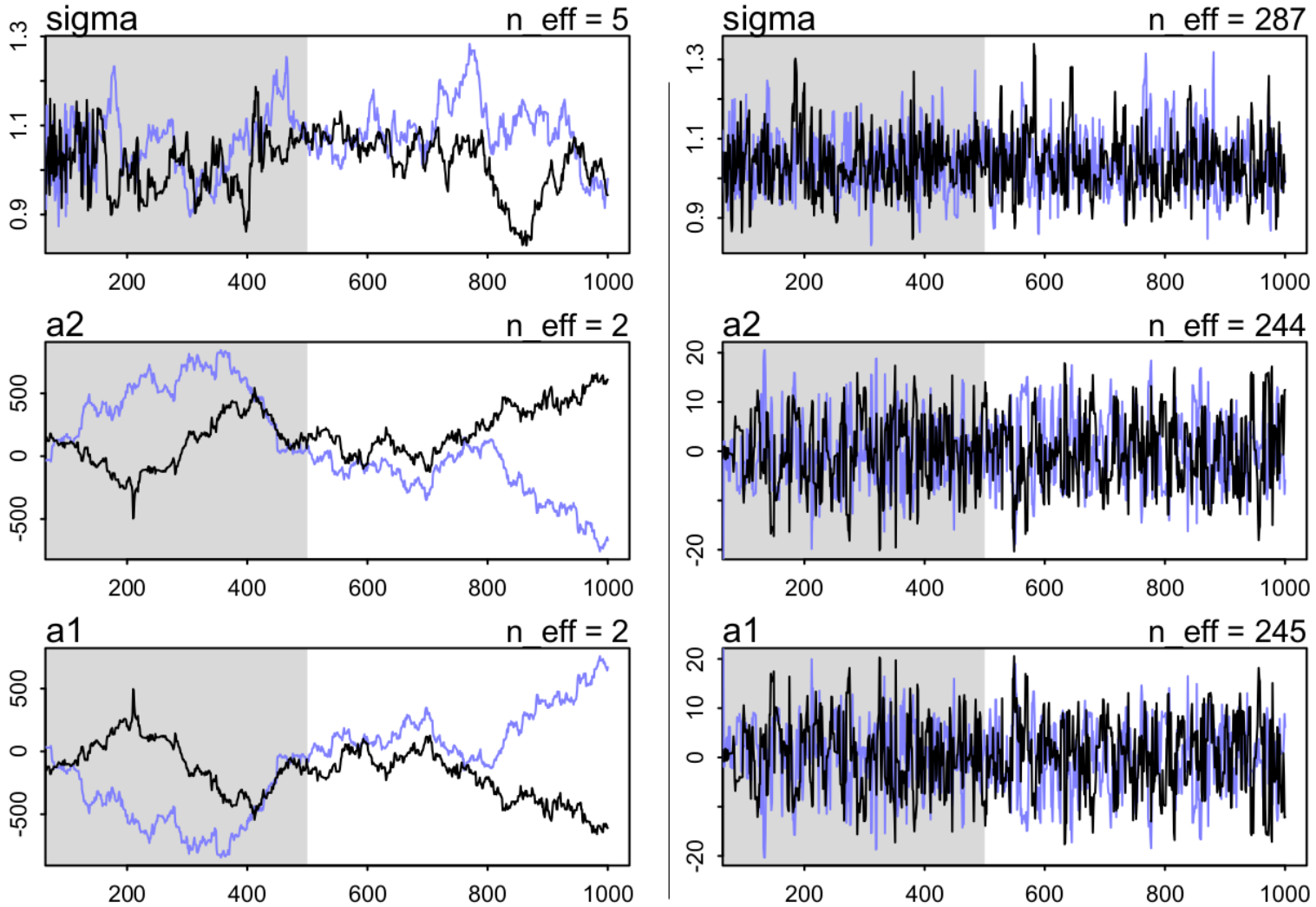
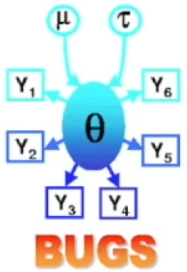


Figure 9.11

MCMC summary

- Desktop MCMC is a recent revolution in scientific computing
- Allows custom scientific modelling



- Allows to deal with large number of parameters in complex models
- Should not be an excuse to stop thinking: GIGA!
  - Measurement errors can get propagated
  - Do not "pipette by mouth"

Checking the Markov owl

A complex machinery, but lots of diagnostics

We have seen that sometimes it doesn't work first time.

- A good chain
  - Converges to the same target distribution
  - Once there, samples efficiently
- Tools for checking
  - Trace plots
  - Trace rank plots
  - R-hat convergence measure
  - Number of effective samples (another convergence measure)
  - Divergent transitions and other special warnings.

What's next?

- Entropy and Generalised Linear Models (GLMs)